

Research Article

Optimal Information Hiding: Advanced Bitstream-Based Watermarking for Secure and Efficient Integration of Image Data in Digital Videos

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Video watermarking is a technique used to embed a visible or invisible mark onto a video file in order to claim ownership, protect intellectual property rights, and detect unauthorized users. The issue of video piracy arises due to the advancement of Internet services and the utilization of various storage technologies. This problem becomes particularly significant with the widespread propagation of media through online channels. The constant progress in sharing of data, videos, images, and applications necessitates the implementation of measures focused on security and the protection of intellectual ownership rights. To address this issue, we proposed a novel Bitstream video watermarking technique that integrates cutting-edge methodologies to seamlessly embed image data within digital videos. This technique stands apart from less sophisticated watermarking methods by capitalizing on spatial and temporal domain concepts in video processing. An experiment is conducted to analyze the time required for the insertion and extraction processes to gain insights into the performance of bitstream watermarking in the video. While enhancing the existing approach, we systematically explored diverse strategies aimed at augmenting watermark capacity, bolstering robustness, and heightening visual imperceptibility. These strategies encompassed the intricate infusion of watermark fragments into the frame area through chaotic mechanisms, the integration of error-correcting codes to ensure data fidelity, and the selective omission of dynamic video segments during information extraction processes. Rigorous experimentations are undertaken by the constraints of lossy compression scenarios, maintaining equivalent or reduced bitrate parameters. The findings substantiate the favorable outcomes stemming from the implemented enhancement, affirming the tangible suitability of the phase watermarking technique for the protection of video copyrights. In order to assess the quality of the bitstream video watermarking technique, the peak signal-to-noise ratio (PSNR), structural similarity index (SSIM), and normalized correlation coefficient (NCC) are computed and examined by the testing data. The suggested model achieved a PSNR value up to 32.52 dB, SSIM value up to 0.97, and NCC value is 1. The dynamic range (DR) and spatial resolution (SR) are measured in the evaluation process. Security is enhanced significantly after implementing the multifaceted approach. The watermarking scheme is more robust and can effectively resist the potential attacks. In summary, this study advances the field of video watermarking by proposing a sophisticated bitstream technique and rigorously evaluating its performance. The outcomes underscore its practical viability for copyright protection, contributing to a more secure digital media landscape.

1. Introduction

The progress in computer and cloud computing technology has made it significantly simpler to produce, save, and distribute digital videos. In the meanwhile, digital video applications have proliferated in people's daily lives due to the Internet's enormous popularity. The digital work property rights of the owner are severely harmed by illegal actions such as piracy, infringement, and tampering [1]. Watermarking is the covert delivery of additional data such as multimedia data with portions of the cover data. Users can easily add information to audiovisual content without altering the original media format. Although the use of watermarking techniques to protect intellectual property rights has recently received considerable attention, this technology has potentially been used across a wider spectrum.

Numerous video watermarking methods have been introduced recently [2]. In the expansive domain of video codec watermarking, a nuanced landscape reveals three distinct methodologies, each embodying unique characteristics, and implications. These primary categories encompass spatial domain video watermarking, watermarking entwined within the encoding process, and the intricate realm of compressed domain video watermarking. It is within this dynamic trichotomy that the underlying mechanisms and challenges unfurl. This dichotomy of watermarking methodologies encapsulates a tapestry of complexities and tradeoffs, where considerations of fidelity, efficiency, and imperceptibility entwine, navigating the delicate balance between securing intellectual property and preserving the integrity of video-encoding processes.

Digital video is extremely important as forensic evidence in court. The capacity to identify tampering is important for video recordings. Watermarking techniques are used to achieve the goals of forensic evidence [3]. The video evidence is not harmed by the watermark, without causing any visual effects. Therefore, the need has been fulfilled by a highly unnoticeable watermark [4]. Digital watermarking techniques were hardly ever associated with image processing. It had been extensively researched for photos, and cutting-edge techniques had been used. There are a lot of brand-new watermarking algorithms available. Several techniques were made for watermarking still photographs, but it seems that the idea has now vanished [5]. Steganography, a method of data concealing, shields confidential information from malevolent attackers during transmission. Good stego image quality, high embedding capacity, low computing cost, visual imperceptibility, undetectability, and increased security are the goals of steganography [6].

The assessment of digital image quality is paramount in various domains, ranging from multimedia content delivery to medical imaging, while traditional metrics such as PSNR and SSIM have been proven invaluable in quantifying image fidelity. They inherently focus on specific aspects of image degradation. This paper proposed an expanded evaluation framework that integrates SR and DR, which addresses comprehensive approach to image quality assessment.

In the rapidly evolving landscape of digital video production and consumption facilitated by advancements in

cloud computing technology, there are concerns surrounding the safeguarding of digital content. Digital video applications are driven by the widespread popularity of the Internet. It has highlighted significant issues such as unauthorized distribution, piracy, and tampering [1]. Watermarking is a pivotal solution to discreetly incorporate supplementary data within multimedia content without altering its original format. The proposed technique not only safeguards intellectual property but also extends its applications beyond protection, potentially encompassing a broader spectrum of use cases. The experimental results show that the suggested method performs well when compared to other methods, as shown in the result section.

Digital video watermarking has unique properties in addition to sharing qualities with digital picture watermarking. The digital watermarking technique concentrates on watermarking images and it has produced numerous scientific findings. PSNR is computed for evaluating the efficiency of the bitstream video watermarking approach. This research paper introduces an extended evaluation framework for digital image quality assessment by incorporating two novel metrics, namely, spatial resolution (SR) and dynamic range (DR). The utilization of these metrics enhances the depth of our assessment methodology, contributing to a more comprehensive understanding of image quality and fidelity [7]. Our contributions hold significant importance due to their impact and value in the following ways:

- (1) Innovative strategies and state-of-the-art advancements are used to embed image data into digital videos.
- (2) Evaluate the quality of the bitstream video watermarking technique by utilizing the peak signal-to-noise ratio (PSNR).
- (3) Digital image qualities are considered by resolution, fidelity of colors, degree of contrast, acuity of details, and the presence of noise artifacts are measured by the dynamic range (DR) and spatial resolution (SR)
- (4) Security is improved by protecting against potential attacks and making it more robust.

In our paper, we have divided the paper into several sections. Section 2 explores the history and background of watermarking. Section 3 reviews related studies in the field. The methodology we utilized is outlined in Section 4, and Section 5 highlights the results we obtained. Lastly, our conclusion can be found in Section 6.

2. Background of Watermarking

Digital watermarking is a method used to encode confidential information into data in order to safeguard the authenticity, authorship, and ownership [8] of the content.

2.1. Digital Watermarking. Digital watermarking revolves around the concept of embedding confidential information to ensure the accuracy, ownership, and originality of the content in a cover media [9].

The process of watermarking is depicted in Figure 1, which involves three distinct steps.

2.1.1. Embedding Stage. In this stage, the embedding algorithm and secret key are used to embed the watermark into the original video. The input includes the video file and the watermark information.

2.1.2. Distortion/Attack Stage. In the middle phase, the watermarked video could be altered by adding noise or exposure to various attacks. The watermarked video in this instance has either been modified or removed.

2.1.3. Detection/Retrieval Stage. At this point, the watermark is extracted from the watermarked video using a detection algorithm and a secret key.

2.2. Properties of Watermarking. The majority of watermarking techniques are used by placing more emphasis on the qualities listed in the following [10].

2.2.1. Error Probability. The practicality of a watermarking scheme is significantly influenced by its error probability [11]. The bit error rate (BER), false-positive rate (FPR), and false-negative rate (FNR) are some significant measures of error probability.

2.2.2. Robustness. The concept of watermarking robustness is often conflated for its application in cryptography [12]. It is important to note that any data that are watermarked can be vulnerable to attacks during transmission. Therefore, it is crucial to ensure that the data remain unaltered, retain its original attributes, and are secured by copyright protection [13].

2.2.3. Capacity and Payload. Capacity refers to the quantity of data that will be encoded in an image. Payload is the collective number of bits in a watermark. The payload may fluctuate from one watermark to the next. Capacity assessment, a crucial aspect of steganography, involves determining the maximum amount of data that can be securely concealed without detection.

2.2.4. Security. The watermark should be able to be extracted by the authorized parties. Hackers or unidentified users should be prevented from accessing the original watermark. The characteristics of the watermark cannot be modified due to potential attacks [14].

2.2.5. Transparency. The watermark is always to be visible to the end user without causing any disturbance to their experience. The consumption of the watermark should not be a hindrance for the user.

2.3. Classification of Watermark. The categorization of a watermark in the main signal is dependent on the method of attachment utilized [15].

2.3.1. Image Watermarking. The method of image watermarking involves embedding confidential information into the image to ensure the author's ownership of an image.

2.3.2. Video Watermarking. This technique embeds confidential data into the video stream to ensure its authenticity and restrict its use.

2.3.3. Graphics Watermarking. Adding a watermark on 2D or 3D graphics to preserve intellectual property is known as graphics watermarking.

2.3.4. Text Watermarking. To maintain the integrity of the original text, confidential information is embedded into the document during the process.

2.3.5. Audio Watermarking. This technique involves encoding sensitive information into the audio channel.

2.4. Watermarking Techniques. This section has discussed various prevalent trends in the practice of watermarking.

2.4.1. DWT. This method of watermarking divides the image into four halves as follows: the approximation, vertical, diagonal, and horizontal components. The image is split into four pieces for the purpose of creating a low-resolution image. The process of calculating the multiple-scale wavelet decomposition involves repeating the procedure. The DWT method is highly favored for watermarking due to its exceptional precision in calculations. The robustness of this technique has a clear advantage in its ability to handle image noise with great effectiveness [16].

2.4.2. DCT. The discrete cosine transform, also known as DCT, has a significant advantage in providing accurate signal approximation by utilizing specific coefficient values. Discrete cosine transformation involves frequency space to represent data [3]. The discrete cosine transformation approach is resistant to basic image processing techniques such as blurring and low-pass filtering. A major advantage of these techniques is that they are relatively quick compared to other procedures.

In this transformation, watermark is implanted on the center frequency bands [17] due to the image breakdown. Compared to previous methods, this approach is resistant to lossy compressions.

2.4.3. DFT. The discrete Fourier transformation is used to break down a continuous function into its frequency components. This technique is commonly used for multibit watermarking. The watermark is incorporated in the first

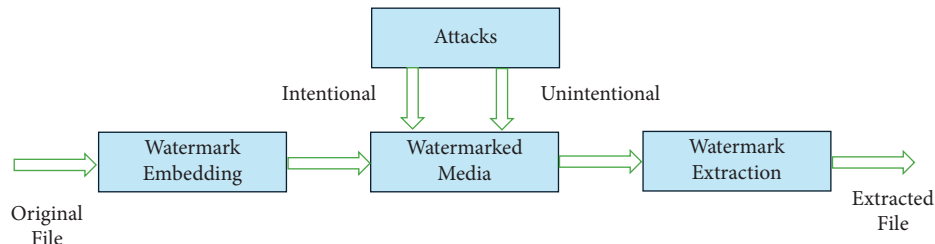


FIGURE 1: Watermarking process.

frames of the collection of images. This method is resistant to geometric assaults, sharpening, and linear and nonlinear filtering [18].

2.4.4. Improved LSB: Adaptive LSB332. The study applied data concealing to raw video files using the LSB332 method, a specific kind of LSB substitution method [19]. LSB332, which is used to conceal eight-bit data to a pixel in the unprocessed video frames, is recommended due to the fact that LSB332 performs better than LSB323 and LSB233 approaches [20].

2.4.5. Fragile Image Watermarking. A high rate of tampering makes fragile watermarking inadequate on the textured background. The researcher wanted to utilize the fragile watermark because it can detect and locate tampered areas, unlike the robust watermark. It uses its visual capabilities to identify edited images and restore them by removing watermarks [21].

2.5. Bitstream Watermarking. The term “bitstream watermarking” refers to the process of embedding directly within the bitstream domain rather than being fully decompressed and recoded [2]. Today, compression technology is widely used to efficiently store and transmit audiovisual content over large networks such as the Internet. This has resulted in an increased need for bitstream watermarking. As a result, bitstream watermarking systems can provide a variety of compelling advantages in the right application circumstances.

- (1) Little complexity in the computation.
- (2) Robustness.
- (3) Control over watermark distortion versus coder distortion.
- (4) Definite quality improvements [22].

3. Related Work

Sun et al. [2] suggested a method that is resistant to recompression, which has no effect on the visual quality of the video and compression ratio. The issue of copyright protection can be resolved via video watermarking technology; it has been the subject of substantial study. In this perspective, research suggests a bitstream-based video watermarking technique based on the recognition of moving

objects [23]. The watermarking system can be made more reliable by using the moving object recognition algorithm to identify the moving macroblocks, which are the part of the moving object in every P frame.

Hua et al. [14] carried out the reversible data hiding in encrypted images using the cipher-feedback secret Sharing technique. In this paper, an RDH-EI system with several data hiders was suggested. Initially, they provide a cipher-feedback secret sharing (CFSS) method that makes use of the secure encryption standard’s cipher-feedback mechanism. A new method with numerous data hiders is then created using the CFSS technique and is known as CFSS-RDHEI. It used an encryption key to split an original image into several smaller encrypted copies, sending each copy to a different data hider [24]. Each marked encrypted image can have its entire original image recovering by using the encryption key.

Li et al. [16] have implemented a reliable video watermarking system capable of withstanding extreme compression. In this research, they have suggested a blind video watermarking mechanism. It can retain integrity on popular social platforms using a processing method that is unknown and has strong robustness to varying compression intensities, and its robustness to varying noise attacks is stable. Furthermore, the algorithm’s computational efficiency is effective and can satisfy real-time needs due to DST’s simple feature. Since extracting watermark information from the receiver does not require any knowledge of the source video, it is more suited for practical applications.

Arab et al. [3] focused on the currently facing difficult issues of video watermarking algorithms, particularly emphasis on robustness of the spatial domain. This study first discusses the problems of state of art, illuminating a paradigm for assessing the effectiveness of video watermarking methods. Additional payload is necessary to incorporate more information to enhance the effectiveness of the strategies. The imperceptibility will be reduced by this additional information. Visual information should not be impacted by the watermark [25]. The framework of this research is to assess the effectiveness of video watermarking methods.

Sangareswari and Aruna [12] suggested and utilized steganography to conceal images in mp4 video. Information-hiding techniques were used to introduce an image covertly into the mp4 video in this work. In addition, extra data can be buried in another frame’s channel, increasing the amount of data that can be concealed. After encoding, the visual quality is practically identical to the

original. In the future work, they will apply steganographic techniques to video files in various formats while testing their stability.

Suresh Kumar and Madhavi Latha [17] proposed the DCT-based secret image obscuration in the video sequence. The main concern is reducing the possibility of information being detected while being sent. The issues are solved by cryptographic methods. To maintain confidentiality of data, it is recommended to hide it since once it is compromised, isolation is no longer effective. One method of digital media can be secured through copyright, which also serves as a mechanism to data concealment. The current LSB modification methodology introduces the risk of losing crucial concealed information. The message bits are distributed at random inside the image. Secret message bits are maintained by higher-order coefficients.

Thakur and Saikia [8] explained the method of data concealing used with AVI picture insertion into video. The solution is tested for 28 frames with the size of 128×128 pixels. A frame can be used in other channels to hide more data that increases the amount of information that can be concealed. It is implemented to explore both the significance of data concealing and the method. The classic current method of LSB modification poses a risk of losing important hidden information. This is because the message bits are randomly distributed throughout the image. The scheme's robustness is not covered in this article [26]. Perceptually, the video's quality after encoding is almost identical to that of the original.

Li et al. [11] introduced secret image-sharing strategies of watermark production. Secret picture sharing can significantly minimize the quantity of data bits encoded in video covers, still ensuring the watermarking's authenticity. The suggested approach enhances the watermarking resistance against various typical signal processing attacks such as noise and recompression while keeping good video quality, according to experimental results.

Ke et al. [26] introduced a novel approach to achieve highly capacitive watermarking of video signals through the utilization of an information mapper. The proposed architecture is centered around the application of deep neural networks. The information mapper plays a pivotal role in facilitating the conversion of a multibit binary signature. A watermark that becomes embedded within the signal frame that is characterized by varying capacitance reflects the entropy measure of the system. The watermark capacity ranges from 4 to 16,384 bits, featuring a resolution of 256×256 pixels. The performance of the algorithms was evaluated using transparency metrics such as the structural similarity index (SSIM) and peak signal-to-noise ratio (PSNR), as well as a robustness metric known as the bit error rate (BER).

The algorithm's robustness is ascertained by its capacity to decode the embedded watermark from the material accurately. The quantification of robustness is accomplished through the utilization of the bit error rate (BER) metric, which involves comparing the original watermark with the

watermark that is extracted from the video [27]. In the realm of watermark embedding and extraction in video files, the following predominant approaches have historically prevailed: spatial domain embedding [28] and transform domain. There are various techniques such as discrete cosine transform (DCT) [29], discrete wavelet transform [30], discrete Fourier transform [31], or a combination of methods [26], as well as compression domain embedding [32]. At present, traditional methodologies are increasingly being supplemented or substituted by the implementation of deep neural network algorithms [33]. It has garnered extensive applications across both civilian and military domains [34]. Neural network algorithms are very efficient, but they require more computational resources, which leads to longer processing times for each frame during the watermark embedding process. This has been noted in research studies.

The practical use of the phase watermarking technique for copyright-protecting videos is described in the study [35]. In place of an error-correcting code model, we compressed a series of frames with an encoded watermark for this study. To increase the extraction quality, we switched from using an actual QR code to using the Reed–Solomon code. In addition, we looked at the video's content and noted large spikes in variance throughout scene changes. When voting for the worth of the extracted code, such frames were disregarded. A conclusion was drawn from the experiment that it is possible to correctly recover the source text from a binary sequence that is embedded in a video with different content. The technique is resistant to noise, light compression, and cropping of the video's opening frames.

Purnima et al. [7] explored the long-term viability of video watermarking approaches through an analysis of watermark and performance evaluations. Video watermarking has developed into a crucial technique to safeguard intellectual property and stop unauthorized use of copyrighted information, considering the development of digital piracy. However, over time, different factors including compression, cropping, and editing can cause the efficiency of watermarking techniques to decline. The purpose of this study is to assess the long-term viability of video watermarking through performance evaluation of various watermarking algorithms and comparison to existing performance evaluators. While several watermarking algorithms demonstrated strong short-term performance, the study discovered that their long-term viability was constrained.

Srivastava and Garg [36] utilized a visual cryptography approach to incorporate the watermark into the cover image following a lot of image processing experiments that produce a better result than other images. Typically, this method evaluates how well an image performs. This method is incredibly helpful for improving image quality and image security in visual cryptography. The outcomes demonstrate the differences in both photos, which nevertheless exhibit the effects of background watermarking and image size. The final product is of higher quality than the original.

Solak and Altinisik [37] proposed a novel two-step data hiding technique, strategically combining image steganography to achieve optimal embedding capacity, enhanced efficiency, improved image quality, and heightened security. In the pursuit of these objectives, we introduced a method that not only bolsters security through encryption with key phrases and a dynamic shifting mechanism but also elevates image quality, capacity, and embedding efficacy. In the initial phase, sensitive data are safeguarded by encrypting it using key phrases and a sophisticated shifting mechanism, forming a robust foundation for security. Subsequently, the technique progresses to the second stage, wherein we introduce the adaptive least-significant-bit (LSB)+3 type I and adaptive LSB+3 type II algorithms to seamlessly embed the encrypted data within the cover image. Empirical assessments affirm the efficacy of the adaptive LSB+3 type I and type II methodologies, demonstrating higher PSNR values in comparison to traditional three-bit LSB substitution methods—reflecting an improvement of 3.48% and 5.73%, respectively.

Zeki Konyar and Solak [20] introduced a streamlined two-stage approach for video steganography, incorporating adaptive LSB32 encoding and advanced Vigenere encryption techniques. In the initial stage, an optimized frame selection process is employed, leveraging the Vigenere encryption method enhanced by the Fibonacci sequence and a three-bit key. The subsequent phase of the proposed methodology entails the utilization of an enhanced adaptive LSB32 method to achieve robust data embedding with heightened capacity.

Loganathan and Ganesan [38] presented an effective reversible video watermarking (Multi-BAM-FUZ) system, which utilises a bidirectional associative memory neural network and HVS based on additional payload or embedding capacity or multiple pictures embedding. Utilising the wavelet approach is more resistant to localized spatial and frequency domain variations due to the wavelet domain. BAM's neural network enables the construction of a weight matrix, which is integrated in the DWT uncorrelated mid, made of numerous images. Frequency coefficients for each Y, Cb, and Cr component of the video frame vary with the embedding factor. Fuzzy inference is utilized to determine the adaptive embedding factor.

Agilandeewari and Ganesan [39] introduced an innovative embedding technique that addresses the mentioned drawbacks by employing a sequential application of the quaternion transform and the conventional curvelet transform. Our approach begins by representing color cover video frames in the quaternion format. Subsequently, we generate coefficients for quaternion curvelet transform (QCT) on each of these quaternion frames. This step is followed by computing the energy of blocks centered around these feature points, enhancing the resilience of the methodology. Next, to uphold the delicate balance between watermarking attributes, we employ the cuckoo search optimization algorithm to discern the optimal embedding positions for the watermark frames. Subsequently, to ensure authenticity, we integrate an authentication key generation process utilizing the owner's biometric image in conjunction with the watermarked frames.

Cong et al. [40] suggested a deep transformer-based network for LF spatial SR dubbed LF-DET. Our architecture consists of two main contributions. One is a convolution layer that is inserted before the self-attention procedure in subsampling spatial modelling to accomplish more affordable and effective global feature extraction in SAI computation. Multiscale angular modelling is the alternative. To address the disparity issue, three parallel branches are used to extract features from micropixel regions with various scales. Results of extensive experiments on numerous open LF datasets show how the approach is superior. In addition, we designed many ablation studies, specifically to verify the advantages, key elements, and design decisions used in LF-DET, including dynamic positional encoding and hierarchical feature aggregation.

Despite the notable success of transformers in various vision tasks, their application to 4D light field (LF) data, which involves larger elements compared to 2D natural images, is still at a relatively nascent stage. Within this context, we introduce a versatile and efficient transformer encoder tailored to LF structures. This encoder demonstrates impressive capabilities in the realm of spatial super-resolution (SR). Moreover, the proposed spatial-angular transformer encoder exhibits adaptability to a spectrum of applications, including object detection and semantic segmentation, domains that have recently converged with LF. This encoder emerges as a favorable choice for feature representation learning. We anticipate that our endeavors will inspire researchers to embrace transformers for the advancement of LF image-processing tasks.

The utilization of Table 1's frequency in this manner leads to a heightened level of precision and superior image quality compared to alternative findings and methodologies presented by other authors.

4. Methodology

According to the literature review, MATLAB is used for the majority of image and video watermarking techniques. The video clip is scaled into frames with the help of a conceptual DCT method. The frame is squared up and can enhance the image-processing experience. It is important to note that this may lead to a slight decrease in visual quality. This processing approach is overhead because the computations are more sophisticated and time consuming. Bitstream video watermarking was selected to address this problem. The insertion of watermark is performed without removing any video frames. The system randomly selects positions from the entire video in order to place a watermark. To implement the proposed method for inserting a bitstream video watermark, we need to follow the following steps.

The process of inserting a watermark is illustrated in Figure 2, and it involves the following steps:

- (1) Choose an MP4 video as the input video.
- (2) The user may select a binary image as a watermark.
- (3) The range of the video file size is used to generate the random integers that determine the random places.

TABLE 1: Comparative analysis of author's methodologies in image watermarking.

Author	Methodology	Strength	Research gap
Saja Abdulamer Kahdim et al., 2023 [41]	Utilize the LWT and IWT methods	To construct the master secret, the IWT was used to convert the concealment images into a frequency domain	The IWT approach is incompatible with any auxiliary watermarking software
Moataz et al., 2022 [42]	Applied the DWT and WHT methods	To solve the noise image issue and defend against assaults	There is a risk that the forged blocks will contain some pixels from neighboring blocks as well
Ernesto Lee et al., 2022 [43]	Approaches for visual cryptography	Analyzed the outcomes of comparing several visual cryptography techniques	It uses a variety of techniques, but it does not specify which is the best
Wu et al., 2021 [6]	Zero-watermarking algorithm	The image is upgraded to stand out from rival zero-watermarking techniques thanks to the high performance	The focus of the research will be on creating multiple color copyright image protection and integrating picture moments
Mehraj et al., 2023 [44]	RBWCI	The proposed robust and blind watermarking-framework (RBWCI) using the discrete cosine transform domain is introduced as an innovative approach for ownership verification and copyright protection	The long-term resilience of the watermarking and encryption techniques against potential attacks or advancements in hacking technologies could be a potential research focus
Chen et al., 2023 [45]	QSVD (quaternion singular value decomposition)	It achieves real-time operation by optimizing complex calculations, indicating its potential for practical applications	It does not provide specific quantitative results or comparisons with existing watermarking methods. A research gap exists in terms of a comprehensive quantitative assessment
Showkat, et al., 2019 [46]	LWT, SVD, and SWT	Experimental results indicate favorable outcomes, particularly in terms of the correlation coefficient of fused watermarks. The technique performs well when compared to various common watermarking schemes, showcasing its potential for high-quality watermarking	Technique could further reduce bandwidth usage by using multiple images for embedding. However, the specific strategies, benefits, and challenges of achieving such bandwidth reduction are not explored
Pandey et al., 2019 [47]	Active video forgery detection techniques	Concept of passive video forgery detection methods, highlighting their potential as an alternative to active video forgery detection techniques. Passive methods are noted for their independence from watermarks or specialized hardware	The future scope does not address potential ethical or legal considerations associated with forgery detection, especially in the context of blockchain technology and data privacy
Srivastava et al., 2019 [36]	DCT and IDCT	The incorporation of discrete cosine transform (DCT) and inverse DCT (IDCT) highlights the use of established image processing techniques for watermark insertion, contributing to the reliability of the proposed method	The feasibility, challenges, and potential limitations of implementing the proposed techniques in real-world scenarios, especially in diverse internet environments, are not addressed
Surekha et al., 2019 [48]	Adding several watermarks	The approach uses extraction and embedding to protect the secret master shares from intrusion	Some attacks, including translations, cropping, scaling, rotations, and contrast alterations, cannot be defended against by this technique

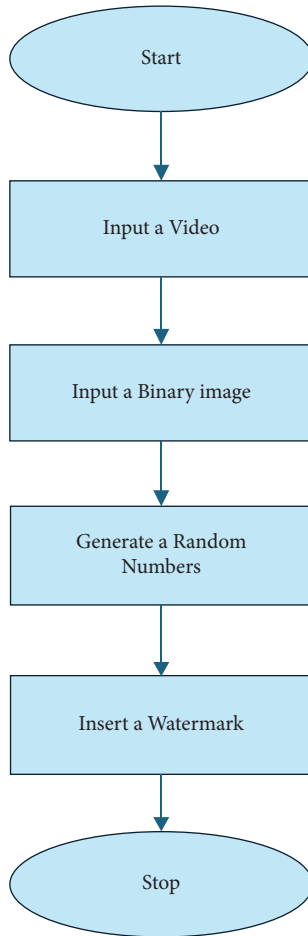


FIGURE 2: Process of inserting a watermark.

- (4) The process of embedding involves placing images in random locations across the video using randomly generated numbers.

It is possible to be compatible with a common video format by choosing an MP4 video as the input. MP4 is a widely recognized format for video compression that enables efficient video transmission and storage. It provides the convenience of embedding various types of MP4 video files with ease.

In Figure 3, the process of extracting watermarks is illustrated as follows:

- (1) A watermarked video is selected as the input video.
- (2) The binary bits are recovered from the relevant random places.
- (3) Compare the extracted binary image with the recovered image.
- (4) Verify the accuracy of the result.

MP4 ensures effective video transmission and storage. A variety of MP4 video files can be easily embedded with this feature. Allowing the user to choose a binary image as the watermark provides them with the freedom and flexibility to customize it according to their preferences. Users have the freedom to select any image, convert it to binary form, and

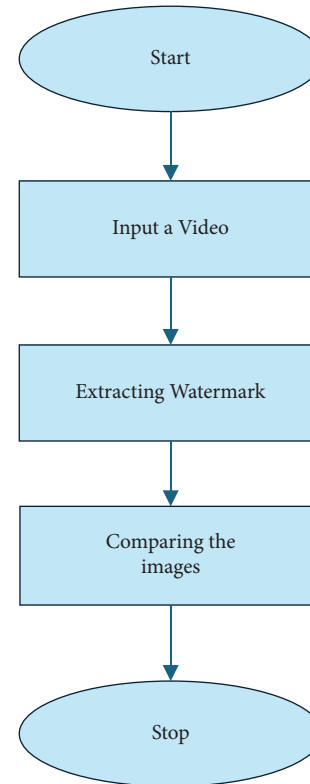


FIGURE 3: The extraction of watermarks.

effectively use it as a watermark. This feature ensures that the watermark in the video is pertinent and significant to the particular context or purpose of the video [49]. The video file's size range is utilized to generate random integers that determine the random placement for the watermark embedding. With this method, the placement of the watermark is guaranteed to be random and unpredictable, making it more difficult for potential infringers or unauthorized users to spot and methodically erase the watermark. The extraction module of the watermark needs the random numbers from the insertion module. The extraction process looks for the embedded data at the given places as specified by the random numbers. The detected watermark will be compared with the original watermark to verify the authenticity of the cover image. If the extracted watermark string does not match with the inserted data, then the test for authenticity will fail.

The security and efficiency of the watermarking technologies are improved by randomization. The binary image watermark is embedded at various points throughout the film using randomly generated numbers. The watermark's security and durability are further increased by the randomization approach. The implementation of a watermark prevents potential attackers from attempting to modify or predict its location in a systematic manner. To maintain the authenticity and ownership of the video content, the random placement of watermarks is generally more effective. In the bitstream method, watermarking is performed without extracting any video frames. The system randomly selects positions from the entire video to place a watermark. The

position of the watermark will be determined by the generated random number. This random number will be the key for extracting the watermark. This is to be shared with the extraction module over a secure channel. The main contribution of the paper is to propose the concept of watermark insertion without the extraction of frames from the video. The paper also proposes the method to insert the watermark at random positions of the video using the bit stream concept. The detailed explanation is given in Algorithm 1.

This stage has several advantages. It supports MP4 videos and enables users to apply their preferred binary graphics as watermarks. In addition, it introduces unpredictability in the watermark placement, which varies based on the video file size. Moreover, it employs randomly generated numbers to insert in random locations. The combination of these characteristics enhances the security, flexibility, and durability of the watermarking method, making it a powerful tool in preventing unauthorized usage or piracy of the video content.

5. Experiment and Results

The experiment was conducted using MATLAB R2023a and an MP4 video file was used as the input. The results were observed for multiple video samples of the same format. The process of embedding a watermark in the video is completed without removing any frames. To apply a watermark, random positions within the video are selected. The entire duration of the video is taken into account while embedding a watermark. Bits from a binary image are embedded into a randomly chosen position. The locations are generated at randomly within the length of the video. The bits are continually implanted at random locations in the cover video after being sequentially read from the image to be embedded.

Insertion and extraction are the two watermarking techniques used to provide bitstream watermarking in a video file. In the process of inserting a watermark, binary image bits are randomly added while the method of extracting the watermark involves retrieving the bits from the binary image.

Random numbers are generated within the range of the video file size. Random numbers can be obtained by utilizing the following random function.

$$R_i + 1 = \{R_i, 0 < R_i < \text{Length of file}\}. \quad (1)$$

Equation (1) denotes the random numbers that falls within the range of the video file size that are indicated by R_i .

The following sequence of randomly generated numbers are represented by $R_i + 1$.

In order to create the watermark, watermarking bits are taken from the image and stored in random locations using generated random numbers.

Watermarking bits are extracted from the image and stored in various random spots determined by randomly generated numbers.

$$C_{iw} = \begin{cases} C_i + 1, & \text{if } B_i = 0, \\ C_i - 1, & \text{if } B_i = 1. \end{cases} \quad (2)$$

In (2), C_{iw} refers to the video's watermark's byte value. The word C_i stands for the original video's byte value. The string's binary values are represented by B_i . When embedding a watermark, the byte value of C_i is increased if $B_i = 0$ and decreased if $B_i = 1$.

During the extraction, it is necessary to reverse the process and reconstruct the image. Insertion and extraction processes outcomes are compared and calculated the processes' execution time. The quality of the inserted and extracted binary images is analyzed.

The binary image of dimension 25×26 is randomly placed within the video and it is recovered from random locations from the video as displayed in Figure 4.

Table 2 lists the execution time for inserting and extracting watermarks from various video files by submitting a binary snapshot with dimensions of 25×26 for watermarking. The table also provides a comparison of execution times for both insertion and extraction processes.

In Figure 5, the execution time for inserting and extracting watermarks from binary pictures of size 25×26 is displayed for various videos. According to the graph, it is observed that the duration for insertion and extraction increases in proportion to the size of the video.

The execution times for the insertion and extraction processes of different binary images provided by the user for watermarking are clearly displayed in Table 3. The comparison made between those two processes is based on the same video file and is highly informative.

Table 3 provides information about the video file "Palm.mp4" and its corresponding image dimensions, insertion time, and extraction time for different resolutions.

The video file "Palm.mp4" is identified and analyzed using insertion and extraction time. Image dimension represents the resolution or size of the images within the video, and it is measured in pixels. The analysis of the video has been conducted at multiple image dimensions, including 650 pixels, 800 pixels, 1100 pixels, and 1500 pixels, as shown in the table. The insertion time refers to the duration taken to encode the video at a specific image dimension, which is measured in seconds. For the "Palm.mp4" video, the insertion times for the respective image dimensions are 47.99 seconds, 57.38 seconds, 63.48 seconds, and 82.05 seconds. The extraction time refers to the amount of time required to extract or decode a video at a particular image dimension. This duration is measured in seconds. In this table, the extraction times for the "Palm.mp4" video at the corresponding image dimensions are 2.31 seconds, 2.54 seconds, 3.02 seconds, and 3.66 seconds.

Figure 6 illustrates the results of the watermark insertion and extraction processes for the same video when the user inputs various binary images for watermarking. Based on the graph, it appears that both insertion and extraction times increase with larger image dimensions.

The proposed method will keep the audio channel intact with the video. The extraction of frames will lead to the loss of audio channel from the video. Separate processing overhead is involved to preserve the audio channel. The strength of watermarking depends on the generation of random numbers. The method has more scope to randomize

```

(1) function VideoWatermarkingAndExtraction (input Video File, output Video File, watermark File, extractedWatermarkFile)
(2)   input Video  $\leftarrow$  Video Reader (input Video File)
(3)   video Path  $\leftarrow$  GetFilePath (input Video)
(4)   file Info  $\leftarrow$  GetFileInfo (video Path)
(5)   videoSizeBytes  $\leftarrow$  GetVideoSize (file Info)
(6)   Display " Video Size:", videoSizeBytes bytes
(7)   output Video  $\leftarrow$  CreateVideoWriter (output Video)
(8)   Set Framerate (output Video, Get Framerate (input video))
(9)   OpenVideoWriter (output Video)
(10)  watermark  $\leftarrow$  Read Image (watermark File)
(11)  resize Watermark  $\leftarrow$  Resize Watermark(watermark)
(12)  watermark Height, watermark Width  $\leftarrow$  Get Dimensions (resize Watermark)
(13)  position X  $\leftarrow$  CalculatePositionX (input Video, width, watermark Width)
(14)  positionY  $\leftarrow$  CalculatePositionY ()
(15)  while Has Frame (input Video) do
(16)    frame  $\leftarrow$  Read Frame (input Video)
(17)    AddWatermarkToFrame (frame, resize Watermark, position X, positionY)
(18)    WriteFrameToOutputVideo (output Video, frame)
(19)  end while
(20)  CloseOutputVideo (output Video)
(21)  Extract Watermark:
(22)  input Video  $\leftarrow$  Video Reader ("output Video")
(23)  frame Number  $\leftarrow$  100
(24)  frame  $\leftarrow$  Read Frame (input Video, frame Number)
(25)    positionX, positionY  $\leftarrow$  GetWatermarkPositionAndSize ()
(26)    extracted Watermark  $\leftarrow$  Extract Watermark (frame, position, positionY)
(27)    extraction Time  $\leftarrow$  StopTimerAndDisplayTime ()
(28) end function

```

ALGORITHM 1: Insertion and extraction algorithm.

FIGURE 4: The binary image input possesses dimensions of 25×26 pixels.TABLE 2: Comparative analysis of insertion and extraction time taken in seconds for different video files and same binary image of dimension 25×26 .

Video	Size (bytes)	Insertion time (sec)	Extraction time (s)
Palm.mp4	38510298	47.99	2.31
MagicTree_2.mp4	44385072	56.88	2.55
Buoy.mp4	60576068	78.23	2.76
Flower_5.mp4	88433719	92.68	2.98

the watermark position as the length of the video increase. This will increase the robustness of the watermark.

Processing time of the algorithm depends on the length of the video as the random number gets generated in

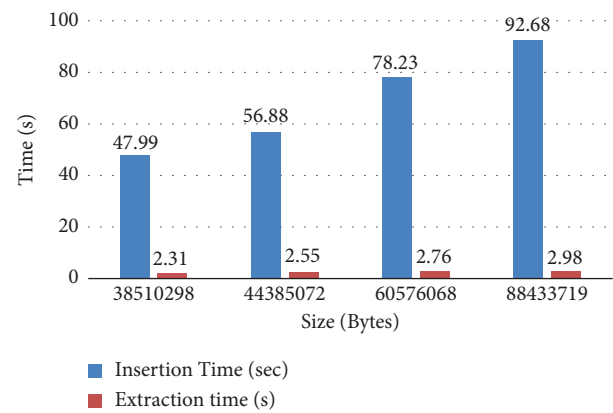


FIGURE 5: Comparative analysis of the insertion and extraction time vs. size of video.

TABLE 3: Comparative analysis of insertion and extraction time taken in milliseconds for same video file and different binary images.

Video	Image dimension	Insertion time (sec)	Extraction time (s)
Palm.mp4	650	47.99	2.31
Palm.mp4	800	57.38	2.54
Palm.mp4	1100	63.48	3.02
Palm.mp4	1500	82.05	3.66

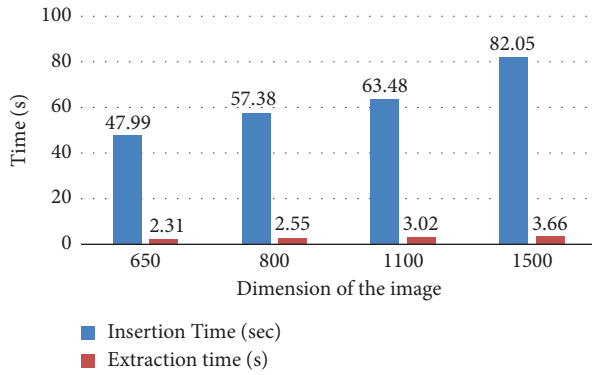


FIGURE 6: Comparative analysis of the insertion and extraction time vs. dimension of image.

proportion to the length of the video to increase the randomness of the number. The programmer is having the option to limit this number to reduce the processing time.

As per the source [10], there is a direct correlation between the quality of videos, images, and the PSNR value, which tends to be higher with better quality. Table 4 lists the PSNR values obtained by computing the PSNR for each frame after extracting frames from both the watermarked and original input videos. Frames are selected randomly and their PSNR is calculated through a comparison of the watermarked and original frames. The PSNR values of the input binary image and the extracted binary image are calculated in a similar manner. A desirable PSNR value is obtained, which remains consistent across the different frames of the films.

The PSNR values of the binary images are observed and compared as follows: Image1.png has a PSNR of 27.52 dB. Image2.png has a slightly higher PSNR of 27.88 dB. Image3.png exhibits a further improvement with a PSNR of 29.52 dB. Finally, the highest PSNR value of 32.52 dB among the binary images is achieved by Image4.png.

Resultant PSNR values can be deduced in that Image4.png has the greatest resemblance to the original video since it obtains the highest PSNR value. This suggests that it preserves visual information better and has less corruption or distortion when compared to the other images. Conversely, Image1.png has the lowest PSNR, suggesting it may have more noise or visual artifacts compared to the other images. Table 4 compares the PSNR values of various binary images extracted from the “Palm.mp4” video, providing a way to assess their relative image quality. The analysis of PSNR values can be found in Figure 7.

In this study, we conducted a comprehensive comparison of the peak signal-to-noise ratio (PSNR) values achieved in previous research works to establish a benchmark for our own methodology. We examined the PSNR results reported as shown in Table 5. By rigorously evaluating our proposed approach, we achieved a notable enhancement in image quality. Our method yielded a significantly improved PSNR value compared to the benchmark set by these prior studies, showcasing the efficacy and advancement of our technique in the domain of binary image processing.”

TABLE 4: Comparison of the PSNR values for the frames of the video and the PSNR values extracted in binary image.

Video	Image name	PSNR of the binary image (dB)
Palm.mp4	Image1.png	27.52
Palm.mp4	Image2.png	27.88
Palm.mp4	Image3.png	29.52
Palm.mp4	Image4.png	32.52

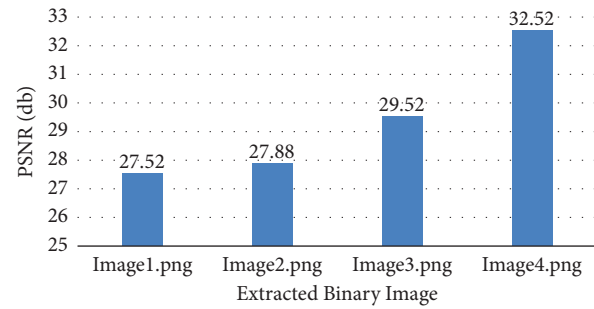


FIGURE 7: Analysis of PSNR values.

To incorporate SR and DR into our evaluation framework, we meticulously calculated these metrics for each image in our dataset. Results are shown in Table 6. SR, representing the density of pixels per unit area, offers insights into the image’s level of detail and clarity. DR, on the other hand, quantifies the range of luminance values captured in the image, contributing to an understanding of its dynamic characteristics. The integration of SR and DR complements the existing metrics, fostering a multidimensional analysis of image quality.

Equations (3) and (4) are used to calculate the spatial resolution (SR) with an image size.

We calculate the total number of pixels in the image as follows:

$$\text{Total pixels} = \text{width} \times \text{height}. \quad (3)$$

We calculate the spatial resolution (SR) as follows:

$$SR = \frac{\text{Total Pixels}}{\text{Area}}. \quad (4)$$

The “spatial resolution” column provides a measure of how densely pixels are packed within the image, given a specific pixel density (PPI) of 150 pixels per inch. A higher SR value indicates that more pixels are present per unit area, suggesting higher image detail and clarity. For example, “Image2.png” has a higher SR (162.22 pixels/sq. inch) compared to “Image1.png” (150.14 pixels/sq. inch), implying that “Image2.png” has a higher level of detail in its composition, as is shown in Figure 8.

Our evaluation demonstrates the significance of including SR and DR in image quality assessment. By considering spatial resolution, we gain a deeper understanding of image detail, facilitating the identification of high-resolution regions critical for various applications. Furthermore, dynamic range analysis reveals the image’s ability

TABLE 5: Results of PSNR employed by each research group in their respective studies.

Author	Methodology	PSNR of the binary image (dB)
Jun Cheng et al. [50]	Segmentation of edges	19.9
Padmashre et al. [51]	LSB algorithm	17.50
Keles et al. [52]	Techniques for compression and restoration	26.68
Tao et al. [53]	Error-controlled lossy compression	20
Ulas et al. [54]	Bit allocation method	28.91

TABLE 6: Evaluation metrics for digital image quality assessment.

Image	Image dimension	Spatial resolution (SR) (in pixels/sq. inch)	Dynamic range (DR)
Image1.png	650	150.14	256
Image2.png	800	162.22	256
Image3.png	1100	176.71	256
Image4.png	1500	182.67	256

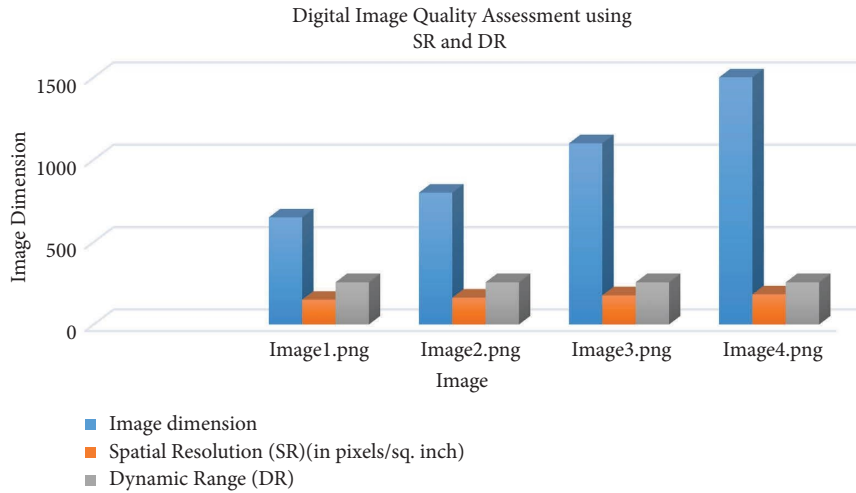


FIGURE 8: Analysis of SR and DR.

to capture and convey a wide spectrum of luminance values. The combination of SR, DR, and PSNR offers a comprehensive depiction of image quality, enabling researchers and practitioners to make more informed decisions regarding image usage and manipulation.

The structural similarity index (SSIM) is another quality evaluation criterion utilized in data hiding [55]. SSIM is a quality criterion that assesses how comparable brightness, contrast, and structural elements are across video or image frames. The following equation represents the calculation of SSIM.

$$SSIM(V_C, V_s) = \frac{(2\mu_c\mu_s + C_1)(2\sigma_{cs} + C_2)}{(\mu_c^2 + \mu_s^2 + C_1)(\sigma_c^2 + \sigma_s^2 + C_2)}, \quad (5)$$

where for cover and stego films, μ_c and μ_s stand for mean and σ_c^2 and σ_s^2 mean variance, respectively. For cover and stego videos, " σ_{cs} " stands for the covariance value. Small constants C_1 and C_2 are dependent on the brightness value. When there is the greatest similarity, SSIM assigns the number 1. The SSIM value decreases as similarity increases and gets closer to zero.

TABLE 7: Comparison of the proposed method with related video steganography techniques.

Method	SSIM	NCC
Proposed	0.97	0.99
Younus et al. [56]	0.86	0
Farri et al. [57]	0.90	1

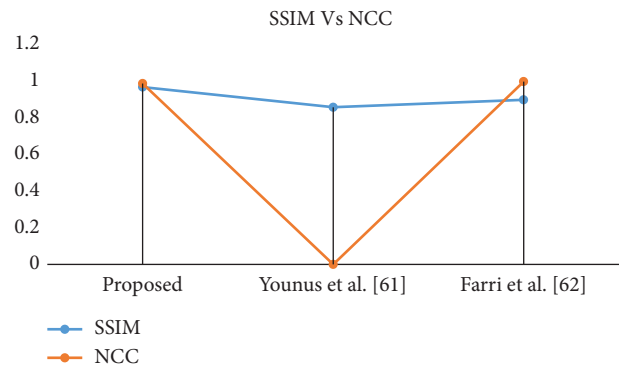


FIGURE 9: Analysis of SSIM and NCC.

TABLE 8: Comparison of the proposed method with related video and image watermarking.

Authors details	Watermarking method	Embedding and extraction process	Robustness against attacks	Evaluation metrics	Practical viability
Our proposed work	Bitstream technique	(1) Chaotic infusion of watermark fragments (2) Integration of error-correcting codes (3) Selective omission of dynamic video segments	Resistance to lossy compression, bitrate constraints	(1) $150.14 < SR < 182.67$ (2) $DR = 256$ (3) $NC = 0.99$ (4) PSNR of the binary image (without extracting the frames) = 32.52 dB (5) $0.94 < SSIM < 0.97$	The proposed bitstream technique demonstrates practical viability by achieving robustness against common attacks, such as lossy compression, and meets essential evaluation metrics for watermarking methods, ensuring high-quality watermarked video with acceptable imperceptibility
		(1) Scrambling watermark using Arnold chaotic map (2) DCT on each medical image block (3) Embedding in DCT middle-band coefficients of each block (4) Extraction based on Weber descriptors without any loss		(1) $0.4167 > NC < 0.97$ (2) $43\text{ dB} < PSNR < 45.93\text{ dB}$ (3) $0.092 < SSIM < 1$	
Soualmi, et al. [58]	DCT + Weber Descriptors + Arnold chaotic map	(1) Generation of binary chaotic sequence (BCS) from a key using algorithm (2) XOR operation between BCS and watermark image for encryption (3) Division of cover image into nonoverlapping 2×2 blocks (4) Generation of CS from a key K (5) Encryption of cover image blocks using CS (6) Embedding encrypted watermark bits into encrypted medical image blocks based on weight and Schur decomposition	Resistant to noising, filtering, and JPEG compression	(1) $0.4167 > NC < 0.97$ (2) $43\text{ dB} < PSNR < 45.93\text{ dB}$ (3) $0.092 < SSIM < 1$	The proposed method shows practical viability by achieving robustness against various attacks commonly encountered in medical imaging. The proposed evaluation metrics indicate acceptable performance in terms of robustness and imperceptibility
Soualmi, et al. [59]	Schur decomposition + chaotic sequence (CS)	(1) Generation of binary chaotic sequence (BCS) from a key using algorithm (2) XOR operation between BCS and watermark image for encryption (3) Division of cover image into nonoverlapping 2×2 blocks (4) Generation of CS from a key K (5) Encryption of cover image blocks using CS (6) Embedding encrypted watermark bits into encrypted medical image blocks based on weight and Schur decomposition	Good image fidelity and acceptable robustness against various attacks	(1) $NC = 1$ (2) $65\text{ dB} < PSNR < 78\text{ dB}$ (3) $0.93 < SSIM < 0.99$	The proposed method demonstrates practical viability with good image fidelity and robustness against various attacks, making it suitable for medical image watermarking applications

TABLE 8: Continued.

Authors details	Watermarking method	Embedding and extraction process	Robustness against attacks	Evaluation metrics	Practical viability
Soualmi, et al. [60]	Imperialistic competitive algorithm (ICA) + speed up robust features (SURFs) + discrete cosine transform (DCT)	(1) Ideal embedding area selected using ICA for optimized cost and complexity (2) Watermark embedded into DCT coefficients of selected areas (3) Blind extraction ensuring color image authentication	Robustness measured by correlation coefficients between original and extracted watermark close to 1 in almost cases	(1) $0.636 < NC < 1$ (2) $61.19 \text{ dB} < PSNR < 63.74 \text{ dB}$ (3) $SSIM = 0.996$	The proposed ICA + SURF + DCT method shows practical viability by achieving high robustness and imperceptibility, making it suitable for applications where secure and authentic color image watermarking is essential
Patro., et al. [61]	Trained with artificial neural networks, multiclass one-against-all support vector machine, and K-nearest neighbors (K-NN)	(1) ECG signals from 20 individuals over 6 months in the MIT-BIH ECG-ID database (2) High-dimensional ($N = 72$) set of ECG features extracted	Least absolute shrinkage and selection operator (LASSO) used to select vital features and prevent overfitting and correlation	(1) Accuracy = 99.139 (LASSO with KNN)	The proposed ECG biometric recognition method effective for security applications where reliable biometric authentication is crucial

The normalized correlation coefficient (NCC) indicates the degree of similarity between two entities are.

The SSIM value of 0.97 indicates that method's output is very similar to the ground truth or reference image. The NCC value of 0.99 suggests a very high level of correlation between the method's output and the reference. In Table 7, it is evident that the proposed method not only achieves the highest SSIM value but also emerges as one of the leading approaches when considering metrics such NCC, which can be observed in Figure 9.

In comparing the proposed bitstream technique with recent works, it stands out for its robustness against common video compression techniques, achieving a high-quality watermarked video. The work in [58] demonstrates effectiveness in medical image watermarking, providing robustness against various attacks. Soualmi's [59] Schur decomposition method excels in maintaining image fidelity while achieving acceptable robustness. The approach in [60] shows promise for secure color image watermarking. Patro's ECG biometric recognition [61] presents a reliable authentication method with a high accuracy of 99.139%. Overall, the choice among these methods should consider specific application requirements and the type of data being processed by comparing the unique aspects and advantages of the proposed bitstream video watermarking technique to emphasize its significance and competitiveness. In contrast to techniques that may lead to a slight decrease in visual quality due to scaling and squaring up frames, our bitstream video watermarking method prioritizes visual fidelity. The watermark insertion is performed without removing any video frames, ensuring a minimal impact on the overall visual experience. The system's ability to randomly select positions from the entire video for watermark placement adds an extra layer of security and unpredictability. This dynamic approach enhances the robustness of the watermarking, making it resilient against various attacks. By avoiding the need to scale and square up frames, our bitstream technique contributes to real-time processing, making it more practical for applications that require quick and efficient watermarking. The versatility of the bitstream approach makes it suitable for different video types, ensuring applicability across a range of multimedia scenarios. The proposed bitstream video watermarking technique combines efficiency, visual quality preservation, dynamic watermark placement, real-time processing, and adaptability, making it a competitive and practical choice for secure and effective video watermarking. Comparison of the proposed method with other related video and image watermarking is shown in Table 8.

6. Conclusion

In this research, we mainly analyzed about bitstream watermarking methods that include both the spatial and temporal domains. Data are watermarked in multiple locations to enhance security. A multilayer defense model is used to prevent unauthorized access or manipulation. Security posture is significantly improved by adopting a multifaceted approach. The process of extracting frames from

videos in this experiment does not impose any additional burden. The time duration of insertion and extraction is based on the size of the video and the dimensions of the binary image.

There are different metrics examined for bitstream watermarking as follows. It is evident from the PSNR values analysis that the bitstream video watermarking technique is more efficient. The accuracy of digital image quality assessment is enhanced by the integration of spatial resolution (SR) and dynamic range (DR). The proposed methods achieve the highest SSIM and NCC values, which outperform other existing models. The research will present a more comprehensive approach to image quality assessment by incorporating SR and DR into the evaluation metrics. This approach proves to be the most practical option for implementing video watermarking in real-world scenarios.

In future, the proposed approach indeed offers a comprehensive and promising methodology for enhancing the security and quality assessment of digital videos through watermarking. The bitstream watermarking technique can be improved by increasing the PSNR, SSIM, and NCC values. The larger size videos are examined and checked by new methods. Potential drawbacks are refined by contributing to the new robust method [62, 63].

Data Availability

All experimental videos are downloaded from the following website: https://all-free-download.com/freefootage/download/palm_6891539.html, https://all-free-download.com/freefootage/download/magic_tree_6891547.html, https://all-free-download.com/freefootage/download/palm_6891539.html.

Conflicts of Interest

The authors declare that they have no conflicts of interest.

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